

II. TIGHT-BINDING APPROACH TO STRAINED KAGOME LATTICE

In this section, we investigate the band structures of strained kagome lattice using the tight-binding approximation. Since the band structures of kagome lattice are primarily characterized by the coexistence of Dirac cones and a flat band, here we mainly focus on the shift of the Dirac cones and the reshaping of the flat band in response to applied uniaxial strain.

A. Tight-binding model for kagome lattice

Fig. 1(a) is an illustration of the nearest-neighbor tight-binding model for strained kagome lattice. The kagome lattice is a triangular Bravais lattice with a unit cell composed of three inequivalent sites, which are labeled as A , B and C , respectively. The three sites in the unit cell form a regular triangle, as indicated by the shaded region in Fig. 1(a). The repeated Brillouin zone of kagome lattice is shown in Fig. 1(b), where the first Brillouin zone is a regular hexagon with two independent corners K and K' .

The tight-binding Hamiltonian of kagome lattice is given as [11, 34]

$$H = - \sum_{\mathbf{r}} [t_1(a_{\mathbf{r}}^\dagger b_{\mathbf{r}} + a_{\mathbf{r}}^\dagger b_{\mathbf{r}+\delta_1}) + t_2(a_{\mathbf{r}}^\dagger c_{\mathbf{r}} + c_{\mathbf{r}}^\dagger a_{\mathbf{r}+\delta_2}) + t_3(b_{\mathbf{r}}^\dagger c_{\mathbf{r}} + c_{\mathbf{r}}^\dagger b_{\mathbf{r}+\delta_3})] + \text{h.c.}, \quad (1)$$

with the summation of $\mathbf{r}$ running over all unit cells. Here $a_{\mathbf{r}}^\dagger$, $b_{\mathbf{r}}^\dagger$ and $c_{\mathbf{r}}^\dagger$ ($a_{\mathbf{r}}$, $b_{\mathbf{r}}$, and $c_{\mathbf{r}}$) are respectively the creation (annihilation) operators of electrons at sites A , B and C in the unit cell located at $\mathbf{r}$, and t_i ($i = 1, 2, 3$) is the hopping parameter along the direction of lattice vectors δ_i , which are given by

$$\delta_1 = ae_x + \sqrt{3}ae_y, \delta_2 = ae_x - \sqrt{3}ae_y, \text{ and } \delta_3 = 2ae_x, \quad (2)$$

where e_x and e_y are basis vectors of the Cartesian coordinate system, and a is the distance between neighboring sites.

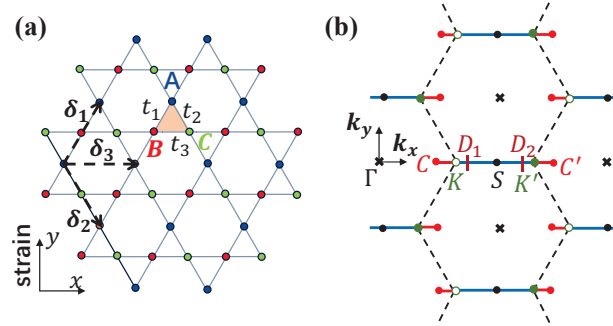


Fig. 1. (color online) (a) Tight-binding model of kagome lattice with only nearest-neighbour hopping terms. The three inequivalent sites in the unit cell are labeled as A , B and C , respectively; t_i ($i = 1, 2, 3$) denotes the hopping parameter between the nearest neighboring atoms along direction δ_i . A uniaxial strain in y direction is modeled by a group of anisotropic hopping parameters $t_1 = t_2 = t'$ and $t_3 = t$. (b) The repeated Brillouin zone of kagome lattice. The center point Γ and two independent corners K and K' are denoted by the black crosses, hollow green circles and solid green circles, respectively. When the lattice is strained in y direction, the Dirac points D_1 and D_2 are shifted away from K and K' and located somewhere between C and C' .

By applying the Wannier transformation

$$\begin{pmatrix} a_{\mathbf{r}} \\ b_{\mathbf{r}} \\ c_{\mathbf{r}} \end{pmatrix} = \sum_{\mathbf{k} \in BZ} \exp(i\mathbf{k} \cdot \mathbf{r}) \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \\ c_{\mathbf{k}} \end{pmatrix} \quad (3)$$

to Eq.(1), the Hamiltonian can be rewritten in momentum space as $H = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \mathcal{H}_{\mathbf{k}} \Psi_{\mathbf{k}}$, where $\Psi_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, c_{\mathbf{k}})^T$ and the Hamiltonian matrix $\mathcal{H}_{\mathbf{k}}$ reads

$$\mathcal{H}_{\mathbf{k}} = \begin{pmatrix} 0 & -t_1(1 + e^{ik_1}) & -t_2(1 + e^{-ik_2}) \\ -t_1(1 + e^{-ik_1}) & 0 & -t_3(1 + e^{-ik_3}) \\ -t_2(1 + e^{ik_2}) & -t_3(1 + e^{ik_3}) & 0 \end{pmatrix}. \quad (4)$$

Here $k_i \equiv \mathbf{k} \cdot \boldsymbol{\delta}_i$ ($i = 1, 2, 3$), which satisfies $k_1 + k_2 = k_3$.

In an unstrained kagome lattice, the 6-fold rotational symmetry ensures $t_1 = t_2 = t_3$. However, as shown in Fig. 1(a), an uniaxial strain applied in y direction breaks the rotational symmetry and consequently leads to $t_1 = t_2 \neq t_3$. Assuming that $t_1 = t_2 = t'$ and $t_3 = t$, the anisotropy ratio α can be defined as $\alpha \equiv t'/t$. The parameter α characterizes both the type and strength of the uniaxial strain, as explained as follows: (i) A decrease in inter-site distance always causes an increment of hopping parameter and *vice versa*, and hence $\alpha > 1$ ($\alpha < 1$) corresponds to the situation where a compressive strain (stretching strain) is applied along y direction. (ii) Apparently, the further α deviates from 1, the stronger the applied strain is.

Meanwhile, as a reasonable approximation for weak strain[23], we neglect the site shift in the strained lattice and assume that the lattice will keep its shape when $\alpha \neq 1$. Based on this assumption, a uniaxial strain applied in y direction can be modeled using a single parameter α , which greatly reduces the mathematical complexity. The eigenvalue equation of $\mathcal{H}_{\mathbf{k}}$ is consequently reduced to

$$E_{\mathbf{k}}^3 + 4\alpha^2 t^3 (1 + \sum_i \cos k_i) - 2E_{\mathbf{k}} t^2 (\cos k_1 + 1) - 2E_{\mathbf{k}} t^2 \alpha^2 (\cos k_2 + \cos k_3 + 2) = 0. \quad (5)$$

B. Results and discussions

1. General features of energy bands

The energy bands of strained kagome lattice can be obtained by solving Eq.(5). Figs. 2(a)-(c) shows the band structures calculated with different values of α .

In the unstrained case where $\alpha = 1.0$, the solutions of Eq.(5) gives two dispersive bands $E_{\mathbf{k}}^{(1,2)} = t[-1 \pm \sqrt{2\sum_i \cos k_i + 3}]$, and a 2D flat band $E_{\mathbf{k}}^{(3)} = 2t$. In resemblance to the energy bands of graphene, the two lower dispersive bands $E_{\mathbf{k}}^{(1)}$ and $E_{\mathbf{k}}^{(2)}$ intersect at the corners of the hexagonal Brillouin zone, forming two linearly dispersive Dirac cones at K and K' , as shown in Fig. 2(a). The flat band $E_{\mathbf{k}}^{(3)}$, which originates from the destructive interference of subdimensional eigen-wavefunctions due to the lattice geometry[13, 34, 39], touches the second band $E_{\mathbf{k}}^{(2)}$ at the center of the Brillouin zone (the Γ point).

The band structures for $\alpha = 0.8$ and $\alpha = 1.2$ were obtained by solving Eq.(5) numerically, as presented in Figs. 1(b)-(c). Similar to the situation in graphene[23], when a uniaxial strain is applied in y direction, the Dirac points of kagome lattice are shifted horizontally away from K and K' . Besides, the applied strain deforms the flat band $E_{\mathbf{k}}^{(3)}$, resulting in a band crossing between $E_{\mathbf{k}}^{(2)}$ and $E_{\mathbf{k}}^{(3)}$.

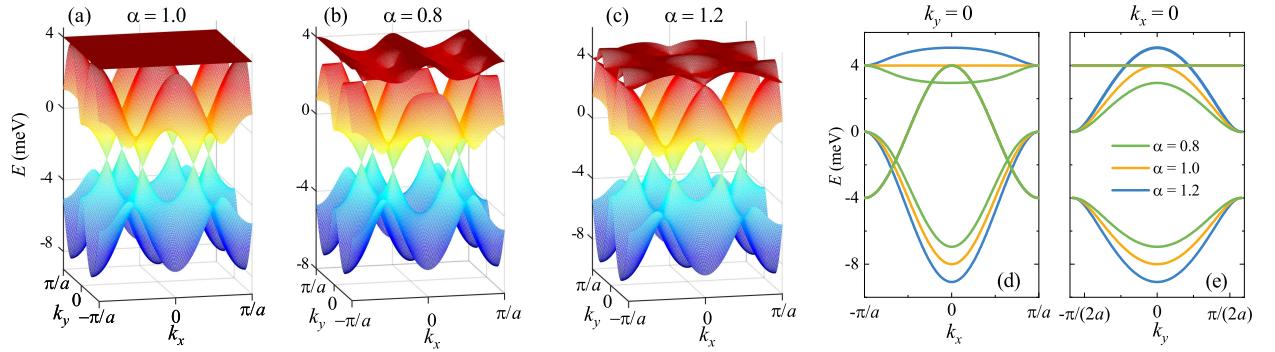


Fig. 2. (Color online) (a)-(c) Band structures of strained kagome lattices with various anisotropy ratio $\alpha \equiv t'/t$, calculated using a tight-binding model with only nearest-neighbour hopping. The lattice is (a) unstrained ($\alpha = 1.0$), (b) stretched ($\alpha = 0.8$), and (c) compressed ($\alpha = 1.2$) along y direction. (d)-(e) Band structures plotted along (d) the k_x axis and (e) the k_y axis. The hopping parameter along x direction is set to $t = t_3 = 2$ meV.

2. Shift of Dirac points

The shift of Dirac points are more visible in the energy bands plotted along the k_x axis. As shown in Fig. 2(d), the intersection points of the two lower bands, i.e., the Dirac points, move with applied strain. The horizontal position